MIL-A-4946B
18 March 1965
SUPERSEDING
MIL-A-4946A(USAF)
03 March 1960

MILITARY SPECIFICATION

ANTENNA ASSEMBLY AT-197( )/GR

This specification is mandatory for use by all Departments and Agencies of the Department of Defense.

1. SCOPE:

1.1 Scope.- This specification covers the requirements for one type of discone antenna assembly 1/ designed to operate at an impedance of 50 ohms, within the frequency range of 225 through 400 megacycles per second (mcs). (See 6.1)

1.2. Classification.-

1.2.1 Type designation (see 6.2).- The type designation of the antenna is derived from the Joint Electronics Type Designation System specified in MIL-STD-196.

2. APPLICABLE DOCUMENTS:

2.1 The following documents, of the issue in effect on date of invitation for bids or requests for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

FEDERAL

PPP-T-60 - Tape: Pressure-Sensitive Adhesive, Waterproof, For Packaging.

PPP-B-636 - Box, Fiberboard.

MILITARY

MIL-C-17/10 - Cable, Radio Frequency, Coaxial, RG-17A/U.

MIL-P-116 - Preservation, Methods of.

FSC 5985

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MIL-C-3643 - Connectors, Coaxial, Radio Frequency, Series HN, and Associated Fittings, General Specification for.

MIL-E-4158 - Electronic Equipment, Ground; General Requirements For.

MIL-F-14072 - Finishes for Ground Signal Equipment.

STANDARDS

FEDERAL

FED-STD-595 - Colors.

MILITARY

MIL-STD-105 - Sampling Procedures and Table for Inspection by Attributes.

MIL-STD-129 - Marking for Shipment and Storage.

MIL-STD-130 - Identification Marking of US Military Property.

MIL-STD-196 - Joint Electronics Type Designation System.

MIL-STD-810 - Environmental Test Methods for Aerospace and Ground Equipment.

DRAWINGS

AIR FORCE

64C3418 - Antenna Assembly, AT-197( )/GR, Discone.

64C3419 - Antenna Mounting.

(Copies of specifications, standards, drawings, and publications required by suppliers in connection with specific procurement functions should be obtained from the procuring activity or as directed by the contracting officer.)

2.2 Other publications.- The following document forms a part of this specification to the extent specified herein.

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Unless otherwise indicated, the issue in effect on date of invitation for bids or request for proposal shall apply.

OFFICIAL CLASSIFICATION COMMITTEE

Uniform Freight Classification Rules.

(Application for copies should be addressed to the Official Classification Committee, One Park Avenue at 33rd Street, New York, New York 10016.)

3. REQUIREMENTS:

3.1 Preproduction.- The antenna furnished under this specification shall be a product which has been tested, and passed the preproduction tests specified in 4.4.

3.2 General service requirements.- The general service requirements of MIL-E-4158 apply as requirements of this specification. Where the requirements of MIL-E-4158 and this specification conflict, the requirements of this specification shall govern. Exceptions and additions to MIL-E-4158 shall be as follows:

3.2.1 Ambient temperature.- The antenna shall be designed to operate over a temperature range of -54°C (-65°F) to +71°C (+160°F).

3.2.2 Relative humidity.- The antenna shall be capable of operation up to a relative humidity of 95 percent between the temperature of 20°C (68°F) and 71°C (160°F) (see 4.6.4.5).

3.2.3 Service conditions (electrical).- The electrical service condition requirements for the antenna shall be as specified herein (see 4.6.2).

3.2.4 Service life.- The antenna shall be designed and constructed for continuous operation with a life expectancy of 5 years.

3.3 Material.- The material for the antenna shall be of a type which will permit the antenna to meet the performance and environmental requirements of this specification.

3.4 Design and construction.- The antenna shall have the design, construction, and physical dimensions specified in AF Drawing 64C3418.

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3.4.1 Polarization.- The antenna shall be designed for vertical polarization.

3.4.2 Antenna mounting.- The antenna mounting shall be as specified in AF Drawing 64C3419 for mounting to antenna mast AB-158/GR.

3.4.3 Input connector.- The input connector shall be of HN femal series and shall have mating dimensions as specified in MIL-C-3643.

3.4.4 Cable.- The cable from the input connector to the radiating elements shall be RG-17A/U in accordance with MIL-C-17/10.

3.4.5 Weight.- The total weight of the antenna shall not exceed 15 pounds.

3.4.6 Finishing.- Before painting, the antenna shall be given a protective finish in accordance with the requirements of MIL-F-14072.

3.4.7 Color.- The color of the paint finish applied to the antenna shall be number 24087 in accordance with FED-STD-595 (see 4.6.1).

3.5 Performance.-

3.5.1 Gain.- The gain, from 225 through 400 mcs, shall not be less than 0.3 decibels (db) over that obtained from a half-wave dipole, tuned to maximum efficiency (see 4.6.3.1).

3.5.2 Voltage standing wave ratio (VSWR).- The VSWR over the entire frequency range shall not exceed 2 to 1 with respect to 50 ohms (see 4.6.3.2).

3.5.3 Impedance.- Normal impedance of the antenna shall be 50 ohms (see 4.6.3.3).

3.5.4 Radiation pattern.- The horizontal radiation pattern shall be omnidirectional and shall not vary from circular by more than 0.5 db (see 4.6.3.4).

3.5.5 Frequency range.- The antenna shall be capable of satisfactory operation without hiatus at any frequency throughout the 225 through 400 mc range. Mechanical tuning to operate the antenna at any specified frequency in this range, after initial manufacturing or factory adjustments, shall not be required (see 4.6.3.5).

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3.5.6 Power capability.- The antenna shall be capable of satisfactory operation with a radio-frequency (RF) carrier input of 1 kilowatt, amplitude modulated 100 percent by a 1,000 cycle per second signal (see 4.6.3.6).

3.6 Environmental.- The antenna shall successfully pass all environmental tests listed in 4.6.4.

3.7 Marking.- The antenna shall be marked in accordance with MIL-STD-130 with the type designation and manufacturer's name, on a nameplate fastened to the antenna tube.

3.8 Workmanship.- Antennas shall be processed in such a manner as to be uniform in quality and all surfaces shall be free of burrs, die marks, chatter marks, scratches, dirt, grease, scale, splinters, and other defects that will affect life, serviceability or appearance.

4. QUALITY ASSURANCE PROVISIONS:

4.1 Responsibility for inspection.- Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Test equipment and inspection facilities.- Test equipment and inspection facilities shall be of sufficient accuracy, quality, and quantity to permit performance of the required inspection. The supplier shall establish calibration of inspection equipment to the satisfaction of the Government.

4.2 Classification of inspection.- The examination and testing of the antennas shall be classified as follows:

(a) Preproduction inspection (see 4.4).

(b) Quality conformance inspection (see 4.5).

1. Inspection of product for delivery (see 4.5.1).

2. Inspection of preparation for delivery (see 4.5.2).

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4.3 Inspection conditions.- Except where specified herein, all inspections shall be made at room ambient temperature, relative humidity, and pressure.

4.3.1 Order of tests.- Visual and mechanical inspection, and VSWR and gain tests shall be performed after power capability and environmental tests have been performed.

4.4 Preproduction inspection.- Preproduction inspection shall be performed by the supplier after award of contract and prior to production, at a location designated by the Government (see 6.2).

4.4.1 Sample.- Three antennas shall be submitted for preproduction inspection.

4.4.2 Test routine.- Sample units shall be subjected to group of examinations and tests listed in Table I for that individual antenna.

TABLE I. Preproduction Inspection

Examination or Test	Requirement Paragraph	Method Paragraph
Antennas 1, 2 and 3		
Visual and Mechanical	3.3, 3.4 to 3.4.7 incl. 3.7, and 3.8	4.6.1
Electrical	3.2.3	4.6.2
Performance	3.5 to 3.5.6 incl.	4.6.3 to 4.6.3.6 incl
Antenna 1		
Rain	3.6	4.6.4.9
Salt fog	3.6	4.6.4.8
Wind	3.6	4.6.4.10.1
Ice	3.6	4.6.4.10.2
Antenna 2		
Fungus	3.6	4.6.4.1
Sand and Dust	3.6	4.6.4.2
Low Pressure	3.6	4.6.4.6
Vibration	3.6	4.6.4.7
Operating dynamic vibration	3.6	4.6.4.11
Antenna 3		
High temperature	3.6	4.6.4.3
Low Temperature	3.6	4.6.4.4
Humidity	3.6	4.6.4.5
Vibration	3.6	4.6.4.7
Operating dynamic vibration	3.6	4.6.4.11

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4.4.3 Failures.- Failure in any of the examinations or tests will be cause for refusal to grant approval of the antennas represented by the samples.

4.5 Quality conformance inspection.-

4.5.1 Inspection of product for delivery.- Inspection of product for delivery shall consist of groups A and B.

4.5.1.1 Inspection lot.- An inspection lot, as far as practicable, shall consist of all the antennas produced under essentially the same conditions and offered for delivery at one time.

4.5.1.2 Rejected lots.- If an inspection lot is rejected, the supplier shall withdraw the lot, and may then rework it to correct the defects, or screen out the defective units. Such lots shall be kept separate from new lots, and shall be clearly identified as lots to be reinspected. Such lot shall be reinspected using tightened inspection.

4.5.1.3 Group A inspection.- Group A inspection shall consist of the examinations and tests specified in Table II and each antenna offered on the contract or order shall be subjected to these tests.

TABLE II. Group A Inspection.

Examination or test	Requirement Paragraph	Method Paragraph
Visual and mechanical	3.3, 3.4 to 3.4.7 incl, 3.7, and 3.8	4.6.1
Electrical	3.2.3	4.6.2
VSWR	3.5.2	4.6.3.2

4.5.1.3.1 Disposition of items.- Items which have passed the Group A inspection shall be subjected to the Group B inspection. Items which do not pass the Group A inspection shall not be acceptable on the contract or order.

4.5.1.4 Group B inspection.- Group B inspection shall consist of the tests specified in Table III.

TABLE III. Group B Inspection

Examination or Test	Requirement Paragraph	Method Paragraph	AQL (% Defective)	
			Major	Minor
Performance	3.5 to 3.5.5 Incl.	4.6.3 to 4.6.3.5 Incl.	1.0	4.0

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4.5.1.4.1 Sampling plan.- Statistical sampling and inspection shall be in accordance with MIL-STD-105 for ordinary inspection. The acceptable quality levels (AQL) shall be as specified in Table III. Major and minor defects shall be as defined in MIL-STD-105.

4.5.1.4.2 Disposition of sample units.- Sample units which have passed the Group B inspection and have no physical or electrical damage, may be delivered on the contract or order if the lot is acceptable.

4.5.2 Inspection of preparation for delivery.- Sample items and packs shall be selected and inspected in accordance with MIL-P-116 to verify conformance with requirements in Section 5 of this specification.

4.6 Methods of examination and test.-

4.6.1 Visual and mechanical examination.- The antenna shall be examined to verify that the design, construction, physical dimensions, weight, finishing, color, input connector, marking, and workmanship are in accordance with the applicable requirements (see 3.3, 3.4 to 3.4.7 inclusive, 3.7, and 3.8).

4.6.2 Electrical test.- Each electrical component shall be given a continuity test to ascertain that it is wired and connected correctly and that good electrical contact is obtained (see 3.2.3).

4.6.3 Performance.-

4.6.3.1 Gain.- The gain of the antenna shall be computed by the comparison of a received signal strength from the test antenna to the received signal strength from a standard half-wave dipole antenna. Measurements shall be made with appropriate test equipment at 225, 310, and 400 mcs (see 3.5.1).

4.6.3.2 VSWR.- The VSWR of the antenna shall be determined by the slotted line method, UHF bridge method, or a method approved by the contracting agency. Measurements shall be made at both extreme frequencies and at least three other evenly spaced frequencies within the operating frequency range. The values shall not exceed that specified (see 3.5.2).

4.6.3.3 Impedance.- The impedance of the antenna shall be measured by one of the methods for measuring VSWR (see 3.5.3).

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4.6.3.4 Radiation pattern.- The radiation pattern shall be determined by the use of a "field-strength" or pattern measuring system, or other method acceptable to the procuring agency (see 3.5.4).

3.6.3.5 Frequency range.- The frequency range requirements shall have been met if the antenna meets the requirements for gain, VSWR and power capability at the frequencies specified (see 3.5.5).

4.6.3.6 Power capability.- Maximum power shall be applied to the antenna for a period of 2 minutes from a transmitter designed to operate with such an antenna, without malfunction of, or damage to, the antenna (see 3.5.6).

4.6.4 Environmental tests.- The antenna shall be subjected to the following environmental conditions in accordance with MIL-STD-810, with the following exceptions, without deterioration of quality or performance (see 3.6).

4.6.4.1 Fungus.- The antenna shall be subjected to the fungus test in accordance with method 508, procedure I.

4.6.4.2 Sand and dust.- The antenna shall be subjected to the sand and dust test in accordance with method 510, procedure I.

4.6.4.3 High temperature.- The antenna shall be subjected to the high temperature test in accordance with method 501, procedure I, except the temperature shall be held at 71°C (160°F) for 4 hours and then at 52°C (126°F) for 20 hours in lieu of 48 hours at 71°C (160°F).

4.6.4.4 Low temperature.- The antenna shall be subjected to the low temperature test in accordance with method 502, procedure I.

4.6.4.5 Humidity.- The antenna shall be subjected to the humidity test in accordance with method 507, procedure I.

4.6.4.6 Low pressure.- The antenna shall be subjected to the low pressure test in accordance with method 500, procedure I.

4.6.4.7 Vibration.- The antenna shall be subjected to the vibration test in accordance with method 514, procedure I.

4.6.4.8 Salt fog.- The antenna shall be subjected to the salt fog test in accordance with method 509, procedure I. After exposure, the salt deposits shall be removed by a

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gentle wash or a dip in running water not warmer than 38°C (100°F) and a light brushing. The antenna shall then be inspected.

4.6.4.9 Rain.- A simulated rainfall equivalent to 4 inches per hour shall be applied to the antenna under the following conditions and in the order specified.

(a) The antenna shall be erected in its normal operating position with the antenna nonoperating.

(b) The first hour the rainfall shall be vertical; the second hour the rainfall shall be 45 degrees to the perpendicular, 15 minutes to a side.

(c) With the simulated rainfall still applied, 1.0 kilowatts of RF power, within the operating frequency range of the antenna, 100 percent amplitude modulated, shall be applied for 2 minutes.

(d) A simulated rainfall shall then be 90 degrees to the perpendicular for 1 hour, 15 minutes to a side, then back to the vertical for 1 hour and the power test repeated.

The performance of the antenna shall not be below the minimum specified herein.

4.6.4.10 Loading tests.-

4.6.4.10.1 Wind.- The antenna, with its normal method of mounting, shall be subjected to direct or simulated loading of wind at 100 miles per hour (mph) for a period of 1 hour. The antenna shall then be examined and shall show no damage.

4.6.4.10.2 Ice.- The antenna shall be loaded with 1 inch of radial ice and subjected to a 60 mph wind for 1 hour. The antenna shall be checked for performance during the test.

4.6.4.11 Operating dynamic vibration.- The antenna shall be subjected to operating dynamic vibration tests across the 225 mc to 400 mc band. With the transmitter at one frequency and the receiver tuned to different frequencies within the band, the antenna shall be shaken such that the total excursion of any part of the antenna shall not be less than 2 inches. When vibrated, the antenna shall not generate or force the transmitter to generate spurious radiation. The purpose of this test is to dynamically simulate wind vibrations as occur under operational environments.

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5. PREPARATION FOR DELIVERY:

5.1 Preservation and packaging.- Preservation and packaging shall be level A or C as specified (see 6.2).

5.1.1 Level A.-

5.1.1.1 Cleaning.- The antenna shall be cleaned in accordance with MIL-P-116, Procedure C-1.

5.1.1.2 Drying.- The antenna shall be dried by one or more of the procedures outlined in MIL-P-116. The procedure(s) selected shall not be injurious to the item.

5.1.1.3 Preservation application.- None required.

5.1.1.4 Unit packaging.- Each antenna shall be preserved in accordance with MIL-P-116, method IIB and packaged in containers conforming to PPP-B-636.

5.1.2 Level C.- Cleaning, drying, preservation, and packaging shall be in accordance with the manufacturer's commercial practice.

5.2 Packing.- Packing shall be level A, B, or C as specified (see 6.2).

5.2.1 Level A.- Antennas packaged as specified in 5.1 shall be packed in fiberboard containers conforming to PPP-B-636, Type I, Class 2. As far as practical, exterior containers shall be of uniform shape and size, and contain identical quantities. The container shall be closed and strapped in accordance with the appendix to PPP-B-636. A case liner is not required when the exterior container is sealed at all joints and seams, including manufacturer's joint, with tape conforming to PPP-T-60, Type III, Class 1.

5.2.2 Level B.- Antennas packaged as specified in 5.1 shall be packed in fiberboard containers conforming to PPP-B-636, Type I, Class 1. Exterior containers shall be of minimum cube and tare consistent with the protection required. As far as practical, exterior containers shall be of uniform shape and size and contain identical quantities. Containers shall be closed and strapped in accordance with the appendix to PPP-B-636.

5.2.3 Level C.- Antennas preserved and packaged as specified in 5.1 shall be packed in a manner to insure carrier acceptance and safe delivery at destination. Containers shall be in accordance with Uniform Freight Classification Rules or regulations of other carriers applicable to the mode of transportation.

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5.3 Marking.- In addition to any special marking required by the contract or oder, unit packages, intermediate packages, and shipping containers shall be marked in accordance with the requirements of MIL-STD-129.

6. NOTES:

6.1 Intended use.- The antenna AT-197( )/GR covered by this specification is intended for use as an antenna for a ground radio transmitter and receiver such as the AN/GRC-27, AN/GRC-29 and AN/TRC-32 for communication with aircraft equipped with UHF command radio sets and such ground UHF stations as necessary in the UHF terminal area.

6.2 Ordering data.- Procurement documents should specify the following:

(a) Title, number and date of this specification.

(b) The complete type designation (see 1.2.1).

(c) Applicable level of preservation, packaging and packing.

(d) Location where preproduction inspection is to be performed (see 4.4).

Custodians:
Air Force - 85
Army - EL

Review Activities:
Air Force - 85
Army - EL, MU

User Activities:
Air Force - 17
Army - MO

Preparing Activity:
Air Force - 85

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SPECIFICATION ANALYSIS SHEET

Form Approved Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity.

SPECIFICATION		
ORGANIZATION	CITY AND STATE	
CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT $

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES" IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

DD FORM 1 APR 63 1426 REPLACES NAVSHIPS FORM 4063, WHICH IS OBSOLETE. AF—WP—O—MAY 64 505

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